2. Task Assignment & Roles

The team consists of five members. All members take part in Flutter/UI implementation, and the remaining responsibility areas are divided among the team. The table below lists each responsibility area with its tasks and output.

Responsibility Area	Tasks	Output
Flutter / UI (all five members)	Work together on implementing the application in Flutter/Dart and integrating the agreed features.	Working Flutter application.
AI / Prompt Engineering	Develop prompt chaining and constrained prompts to generate relevant scenarios and level-appropriate dialogues that use known vocabulary while introducing relevant expressions.	Prompt flow and reviewed AI-generated dialogues.
State Management	Implement Cubit/BLoC state handling for active conversations and learning progress.	Structured state handling for conversations and progress.
Local Data Storage	Use Hive to store learned vocabulary, learning progress, and selected lesson snippets for offline access.	Persistent learning data and saved-content access.
Security & Data Protection	Protect locally stored learning data and cryptographic keys through encryption and the Android Keystore.	Data-protection mechanisms aligned with the proposal.
Testing & Final Documentation	Test core user journeys, AI output suitability, storage, offline behavior, and error handling; contribute to the final report and presentation.	Test results, final report, and presentation materials.

Flutter / UI work is shared by all five members. The other responsibility areas are distributed among the team members.

3. Risk Assessment & Mitigation Plan

The table lists the main risks identified from the planned features. Likelihood and impact are rated High, Medium, or Low as the team's working estimate, and the ratings are updated as the project progresses.

Risk	Likelihood	Impact	Mitigation / Prevention	Contingency / Response
AI dialogue does not match the learner's proficiency or uses too many unfamiliar words.	Medium	High	Use learner context, constrained prompts, and known vocabulary; review sample outputs across proficiency levels.	Revise the prompts and add output review and retry behavior.
AI response is slow or the service is unavailable.	Medium	High	Handle loading, timeout, and error states; test under different network conditions.	Explain the issue to the learner and keep previously saved content available offline.
AI API key is exposed or usage limits are reached.	Medium	High	Keep the key out of source code and version control; monitor usage during development and testing.	Replace the key if exposed; fall back to saved content until limits reset.
AI integration or feature development takes longer than expected.	Medium	High	Test the AI request/response flow early and prioritize the features in the initial scope.	Focus on the core text-based conversation flow and defer extra enhancements.

Risk	Likelihood	Impact	Mitigation / Prevention	Contingency / Response
Saved vocabulary or progress is not stored or restored correctly.	Low	High	Test Hive read/write behavior, app restarts, and updates to stored data.	Fix persistence defects and retest before the final demonstration.
Saved content cannot be accessed offline.	Low	Medium	Separate internet-dependent AI features from saved local content; test with internet disabled.	Limit offline functionality to saved vocabulary and lesson snippets, as in the proposal.
Learner data or cryptographic keys are inadequately protected.	Low	High	Minimize stored learner data and protect it with encryption and the Android Keystore.	Fix high-priority security issues before the final presentation.
Missed deadlines or integration conflicts.	Medium	Medium	Divide tasks clearly, agree on interfaces, use version control, and review progress against the DEPI dates.	Reprioritize in-scope work and communicate schedule changes to the team.

Risk monitoring: The team reviews these risks during check-ins and updates them as the design and implementation progress.

4. KPIs (Key Performance Indicators)

The KPIs below measure the success of the initial release. Targets are set once the team has agreed on test conditions and recorded baseline results.

KPI	How to Measure	Target
AI Response Time	Measure the time between sending a request and receiving a usable dialogue across a documented set of test requests.	Record a baseline, then set the response-time target from it.
Dialogue Relevance & Level Suitability	Review generated dialogues for scenario relevance, fit with the selected proficiency level, use of known vocabulary, and limited introduction of new expressions.	Set a review checklist and acceptance threshold before final testing.
Core Flow Success Rate	Track test runs that complete learner setup, scenario generation, text conversation, and saving without a blocking error.	Set the required pass rate once the test cases are defined.
Crash / Error Rate	Record crashes and blocking errors during repeatable test sessions.	Set the acceptable rate from test results and release needs.
Local Data Persistence	Check that saved vocabulary, progress, and lesson snippets remain available after restarting the app.	All critical persistence test cases pass.
Offline Saved-Content Access	Test that saved vocabulary and lesson snippets open with internet access disabled.	All critical offline test cases pass.
On-Time Task Completion	Compare tasks completed by their DEPI deadlines with tasks planned for those deadlines.	Track completion against the DEPI schedule and document delays.

These indicators cover the initial English, text-based release. Voice conversations, pronunciation scoring, additional languages, and fully offline AI are outside this release.